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COGNITREK
Driving
Cognitive
Accessibility
NGI SEARCH

COGNITREK

Analysis and Architecture

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1. COGNITREK Project

1.1. Introduction

The present document outlines the proposed Architecture for the COGNITREK project after reviewing the Requirements proposed for the Platform, as well as the possible Use Cases, both complementary from a technical level and derived from the specifications of the Business Plan.

The document is structured as follows: Project Description and Objectives, Requirements Analysis, Actors and Use Cases, Architecture, and Software Components. The Appendices include references to comparisons between evaluated LLM systems, bibliographic references, etc.

1.2. Project Definition

COGNITREK is an innovative initiative focused on creating a system for individuals to understand complex documents via a conversational interface.

Going beyond classic web accessibility, we aim to enhance cognitive accessibility, breaking down barriers to understanding information for a more inclusive digital experience. The Ximdex platform's AI-driven systems have already automated digital content to provide web accessibility through, among others, relevant entity recognition, image captioning, and text translation. Yet, a persistent challenge remains—adapting text to individual readers, particularly in educational settings.

COGNITREK's primary objective is Cognitive Accessibility, achievable through Decoupling Information, Providing Context, and AI-based Conversation and Dialogue. Our mission is to create a seamlessly inclusive digital landscape, revolutionizing the way people access and comprehend information.

2. Requirement Analysis

The development of a visual system for Educational Content Adaptation has the following Functional and Non-Functional Requirements:

- FR1: User Management [XDIR Module]
- FR2: Content Viewing [VISOR Module]
- FR3: Contextual Viewing of Content Adaptation [VISOR]
- FR4: Generate New Adaptations [TRANSFORM Module]
- FR5: Evaluation of Adapted Content [VISOR]
- FR6: Sharing Adaptations with a Student [VISOR]
- FR7: Adaptation Record for a Student [VISOR]
- FR8: Storage of Completed Adaptations [XDAM]
- FR9: Profile Storage [PROFILE Module]

2.1. Requirements derived from the Ethics Analysis

Se recogen a continuación una serie de requisitos provenientes del Análisis del Proyecto desde el punto de vista Ético indicando a continuación que Requisito (RF) lo tendrán en consideración:

- Introduce mechanisms for human oversight (teachers, tutors, parents, development team) on decisions made by AI to assess and eliminate potential biases and to maximize the student's intellectual development → FR3.
- EthReq2: Facilitate Adaptive learning processes with self-regulation through student evaluation (OK / NOK) of the offered outcome → FR5.
- EthReq3: Provide students with contexts of work previously supervised by the tutor as for example: a cooperative system where the teacher prepares a context and shares it with the student → FR6.
- EthReq4: Consider GDPR and related regulations with special attention to data, especially sensitive since the users are minors → FR1, FR9.
- EthReq5: Compliance with the “Requirements for child-centred AI” by UNICEF (pg. 31), with special attention to the following:

    - 1: Supports children’s development,
    - 3: Fairness and non-discrimination,
    - 4: Privacy,
    - 5: Safety,
    - 6: Transparency, accountability and explainability → FR7,
    - 7: Knowledge of their rights,
    - 9: Create an enabling environment → FR9.

- EthReq6: Minimize stored data (age, educational level, and cognitive adaptation performed), omitting other socioeconomic, ethnic, or biometric data, and ensure extreme security, especially for sensitive data → FR1, FR9.
- EthReq7: Provide systems in a protected "closed cage" mode to prevent free access to prompts to avoid misuse by the student or third parties.
- EthReq8: Use LLMs systems at a low level (i.e., in the case of OpenAI, do not directly use ChatGPT but work with OpenAI libraries providing embeddings) to avoid "hallucinations" and out-of-context responses.
- EthReq9: Quickly rotate stored information if background analyses (dialogs, contexts, scoring) are performed.

The Ethical Requirements EthReq 5, 7, 8, 9 will be considered transversally by the COGNITREK project Architecture.

3. Relevant Entities y Actors

The main entities of COGNITREK are:

- **Users**: system users (students, teachers) who will have access to differentiated functionalities.
- **Content** (adapted or not): educational resources in SCORM or PDF formats.
- **Adaptation Filters**: a simple task of adapting content for a specific purpose (e.g., dyslexia).
- **Adaptation**: a complex flow of simple adaptations to obtain content fully tailored to the student's needs.
- **Evaluations**: regarding the quality of the adaptations, to be shared with the tutor.

The actors involved in the use of COGNITREK are:

- **Administrator**: responsible for configurations, non-automatic registrations, etc.
- **Tutor** (teacher, parent): responsible for proposing adaptations for students and reviewing their quality, both through direct viewing and student feedback. They can activate adaptations upon the student’s request.
- **Student**: the final recipient of the adapted content. They can review both original and adapted content and provide feedback in the form of evaluations.

4. Use Cases

UC-1	*Login*	
[Version]	*1.0*	
[Dependencies]	*None*	
Precondition	*The user is not logged in and has access credentials to the system.*	
Description	*The system should behave as described in the following use case when a user attempts to log in.*	
Sequence	**Step**	**Action**
	1	*The user enters their username and password.*
	2	*The system verifies that the credentials are correct and responds by either granting access to the Main Screen (Step 3a) or displaying an error and returning to the Login Screen (Step 3b).*
	3a	*The user gains access to the main screen if the credentials are correct.*
	3b	*The user is returned to the login screen if the credentials are not recognized.*

UC-2	*Logout*	
[Version]	*1.0*	
[Dependencies]	*None*	
Precondition	*The user is logged into the system.*	
Description	*The system should behave as described in the following use case when a user attempts to log out.*	
Sequence	**Step**	**Action**
	1	*The user clicks on the logout button.*
	2	*The system logs the user out and redirects them to the home page.*

UC-3	*Gestión del Usuario*
[Version]	*1.0*
[Dependencies]	*None*
Precondition	*The user is logged into the system.*
Description	*The system allows the user to modify the minimum required data as defined by the COGNITREK system, unsubscribe, or regenerate their access credentials.*

UC-4	*Launch and Share Adaptation Sequence*	
[Version]	*1.0*	
[Dependencies]	*None*	
Precondition	*The Tutor or Administrator user is logged into the system.*	
Description	*The privileged user (tutor or administrator), from the screen of an educational content in XDAM initiates an adaptation sequence for a student.*	
Sequence	**Step**	**Action**
	1	*The user clicks on "Choose Student and Adaptation to Perform."*
	2	*The Adaptation Generator processes the request and uploads it to XDAM.*

UC-5	*Editing an Adaptation Sequence for a Student*
[Version]	*1.0*
[Dependencies]	*None*
Precondition	*The user is logged into the system and is either a Tutor or an Administrator.*
Description	*The system displays the steps to be applied to adapt educational content (e.g., for dyslexia or content simplification) and allows the user to reorder, delete, or add new steps.*

UC-6	*View Educational Content (Original and Adapted)*
[Version]	*1.0*
[Dependencies]	*None*
Precondition	*The user is logged into the system.*
Description	*The system displays an educational content resource (in PDF or SCORM format) and provides access to view the original version, the adapted version, consult the applied adaptation sequence, and evaluate the quality of the adaptation.*

UC-7	*Evaluate Adapted Educational Content*
[Version]	*1.0*
[Dependencies]	*None*
Precondition	*The user is logged into the system and is viewing adapted content.*
Description	*The system displays the adapted educational content (in PDF or SCORM format) and provides access to view the original version, the adapted version, consult the applied adaptation sequence, and presents a form to evaluate the quality of the adaptation by adding a comment. This feedback can be accessed by the Tutor and Administrator to assess the quality of COGNITREK.*

5. Architecture and Software Components

Decoupled architecture based on services and components using the Ximdex Platform (XDIR and XDAM) as well as other necessary components.

With Graphical Interface:

- XDIR: User management system integrated with the XIMDEX Platform.
- XDAM: Digital resource management system with a search engine and workflows.
- VISOR: System for viewing Educational Resources (PDF, SCORM). Based on the user role, it will provide access to various functionalities, such as launching an adaptation for a student and suggesting it to the student accessible by tutors, evaluating the adaptation for the student, etc.

Without Graphical Interface:

- TRANSFORM: Adaptation Sequence Generator, a component based on LLMs to perform adaptations in the form of sequential flows.

5.1. Infrastructure

The system will primarily use PHP, Python, JavaScript, CSS, and HTML5.

Documentary data will be stored in XDAM, which in turn uses MySQL (or compatible) and Apache Solr (for searches).

Relational data will be stored in project-specific databases, ensuring compliance with the aforementioned Ethical Requirements.

The adaptation systems will attempt to directly execute LLMs. If this is not feasible, the use will be documented, evaluating the ethical impact. All components will run on Linux operating systems on European servers, without sharing information with third parties except for documents to be adapted using LLMs.

El sistema usará principalmente PHP, PYTHON, JS, CSS y HTML5.

Los datos documentales se almacenarán en XDAM que utiliza a su vez MYSQL (o compatible) y Apache SOLR (para las búsquedas)

Los datos relacionales se almacenarán en BBDD específicas del proyecto, velándose por los Requisitos Éticos mencionados.

The adaptation systems will attempt to directly execute LLMs. If this is not feasible, their use will be documented, evaluating the ethical impact. All components will run on Linux operating systems on European servers, without sharing information with third parties except for documents to be adapted using LLMs.

5.2. Software Component Description

The four basic components in the COGNITREK Architecture are:

5.2.1. Digital Asset Management, XDAM

The Digital Asset Management service of the Ximdex Platform (XDAM) provides a single point of interaction with all types of digital objects (images, videos, audio, e-books, documents, fonts, links, websites, etc.), categorizing them for quick and efficient search based on customizable facets.

The following image illustrates the interface of the XDAM service, displaying resources related to a specific search as well as the facets associated with the use of those resources by other services:

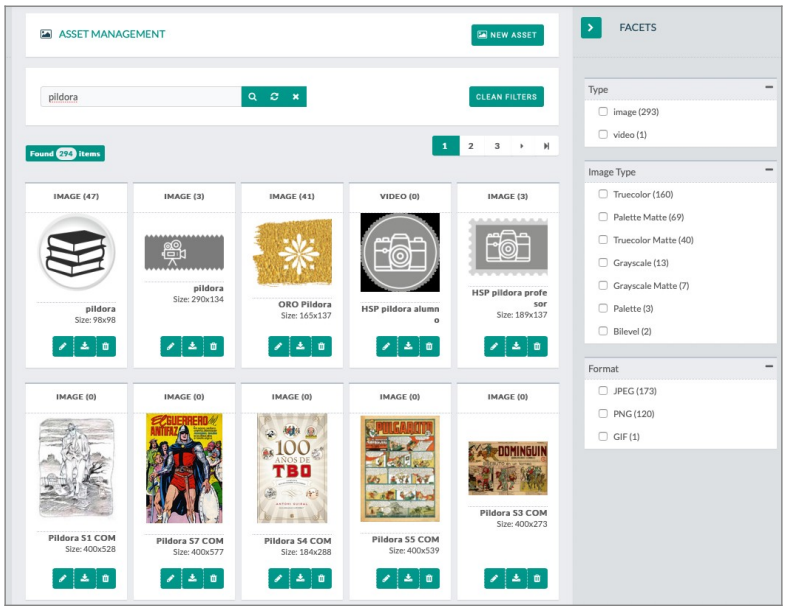


Fig. 1: A possible graphical interface of the XDAM service showing filtering facets.

The gallery will display all resources that we have access to view, classified by type, keywords, etc.

Support for role and user group management allows control over viewing capabilities (private, public, by groups) and editing of resources and their metadata.

The repository also enables the uploading of new resources and editing of existing resources and their associated metadata. Depending on the permissions granted, each resource will display a series of buttons to perform actions such as editing, downloading, or selecting it to add to a collection.

XDAM also includes metadata editors (schema.org, LOM-ES, etc.) to customize the metadata associated with resources and manage them as Educational Digital Objects (ODEs).

XDAM is essentially an aggregator of digital repositories with support for other remote repositories, file storage systems (Google Drive, Dropbox), or content management systems like XCMS, Drupal, etc.

The following figure shows the editing screen for an Educational Digital Object and its metadata in LOM-ES format:

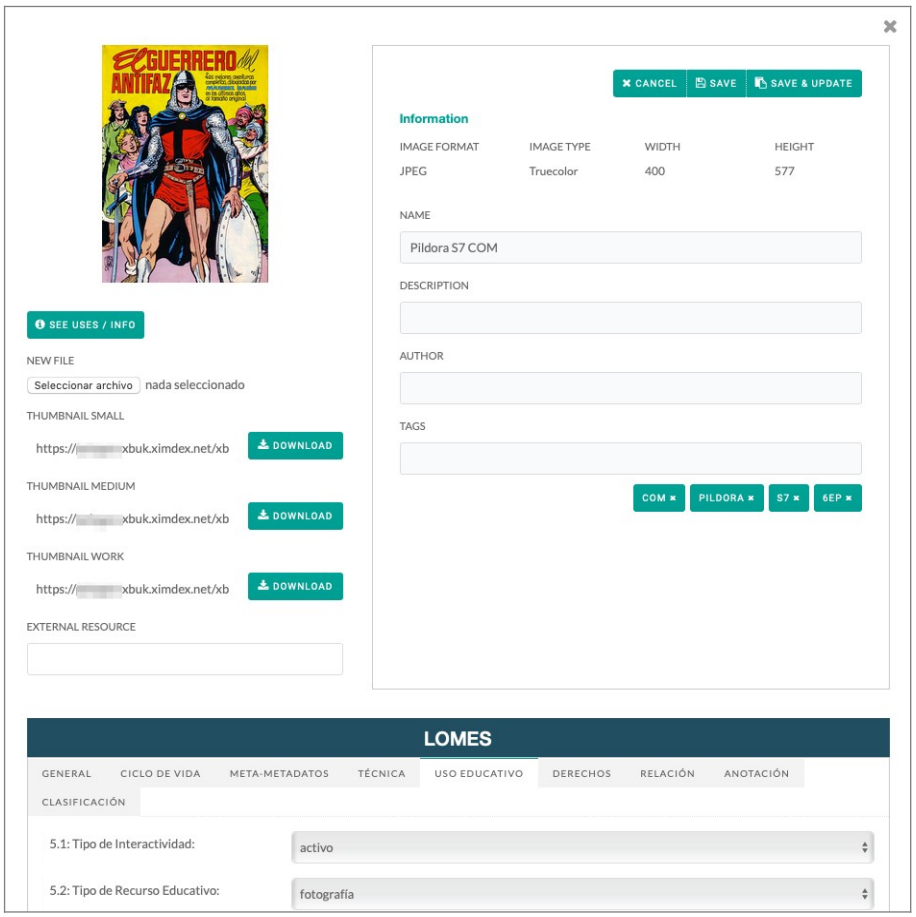


Fig. 2: Editing a resource in the repository including its LOM-ES metadata.

Use of XDAM in COGNITREK

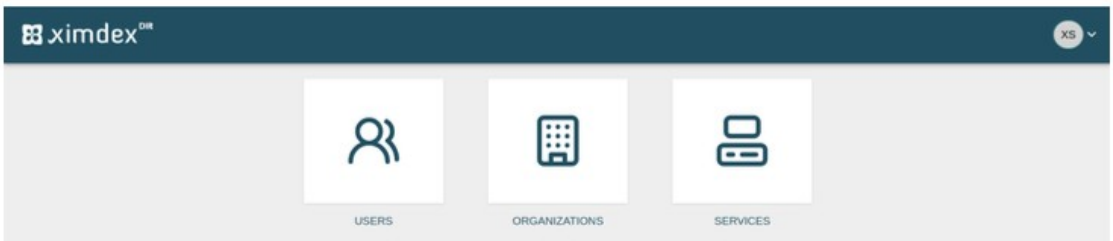
The Digital Asset Management developed by XIMDEX and published under a free license in the context of the COGNITREK project will be used as a document resource manager. It provides the following functionalities:

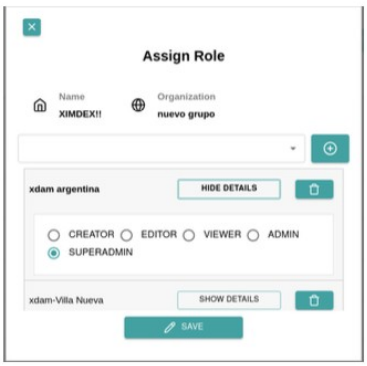
- Management of Organizations
- Structured in Collections based on types of digital resources stored (courses, books, multimedia, etc.)
- With specific metadata by collection type, including default LOM metadata.
- Allows multiple files to be associated with the same resource in a collection.
- With Content Delivery Network functionality, creating a link (protected or not) to the resource for sharing in viewers, other systems, etc.
- With Transcoding functionalities to automatically generate lightweight versions of videos or images.
- With an associated workflow for each type of resource. In COGNITREK, it will be used to provide access to adaptation sequences for the Tutor.

5.2.2. User and Organization Directory, XDIR

The XIMDEX platform previously managed users without unification, with each component using its own strategy. Within the framework of the COGNITREK project, we recognized the need to simplify this aspect, leading to the development of a comprehensive user management system (registration, deactivation, and editing; credential recovery). This system also allows for the association of roles and organizations, as well as services within the XIMDEX Platform.

The following figures illustrate the administration panel and role assignment:



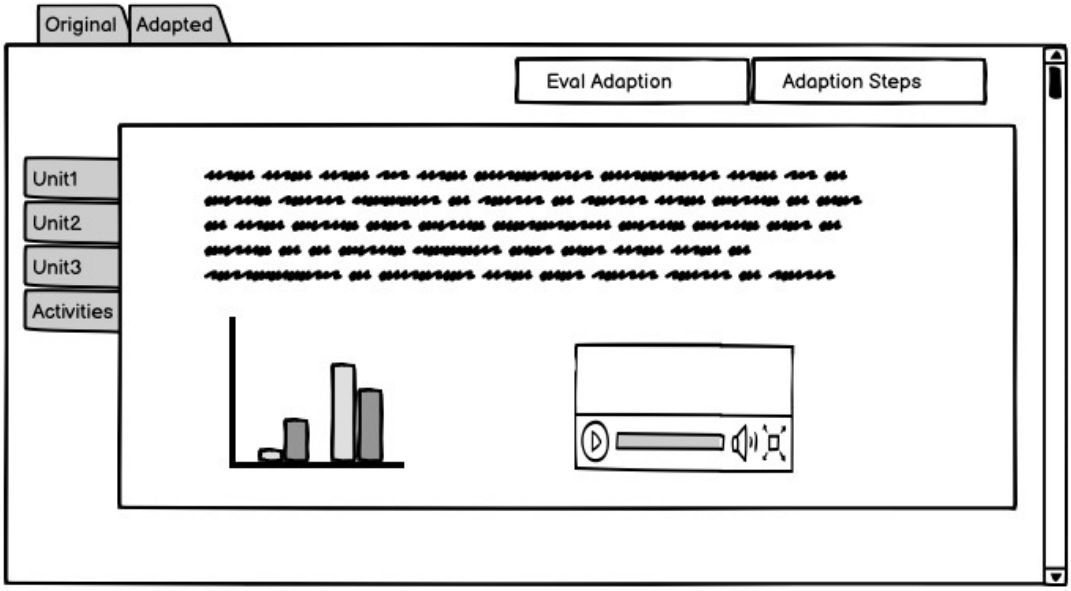


5.2.3. Educational Resources visualizer, VISOR

The key point of COGNITREK is the graphical interface used by students to view both the original content and the content adapted by the system.

A web viewer is proposed for documents in PDF format and educational content in SCORM format. In addition to providing access to the content, it will offer information about the adaptation sequence followed and allow users to evaluate the quality of the generated material to provide feedback to both the Tutor and the developers.

Below is a mockup showcasing the possible appearance of the Viewer from the Student role:



5.2.4. Transformer to apply a Sequence of Adaptations, TRANSFORM

The COGNITREK system will allow each student to be associated with an adaptation profile that automates the process of adapting content for each student.

The advantages of this approach are varied:

- It allows for the completion of time-consuming transformation tasks so that the student can access them alongside the original content.
- It enables the Tutor to preview the adaptations made to modify and decide whether to share them with the student.

- It facilitates traceability and evaluation of the adapted content.
- It allows for the grouping of adaptations made for use by multiple students.

A key point in the process is Student Profiling to associate an adaptation sequence with each student. Regarding this aspect, it is not yet predetermined whether this will be done automatically, semi-automatically, or completely manually by the Tutor. Initially, we lean towards making a sequencing proposal, which the tutor can then finalize or provide optional steps for the student.

Some of the steps considered in the adaptation sequence are to support:

- Dyslexia.
- Reading comprehension.
- Color blindness and vision problems.
- Attention deficit.
- Auditory or Visual learning styles (optional adaptation to be chosen by the student).

Multiple LLM systems with very different characteristics have been evaluated. The following appendix includes metrics on behavior and performance based on the complete comparative appendix titled “”

During the development of the upcoming milestones of the COGNITREK project, one of the LLMs (or several if necessary) will be selected to perform the various adaptation tasks listed, first considering the need for it to run locally and, if unavoidable (due to performance or licensing issues), that it runs remotely, ensuring it has no access to personal data and that the server and service operate within the EU.

6. Summary Sheet of Evaluated LLMs

A set of LLMs has been evaluated in the context of the COGNITREK Project. The annex titled "20240925_COGNITREK_Analysis_ANNEX-LLMs-ES.pdf" contains a comparison that considers cost, technical aspects, deployment, and more.

The following graphs present a quantitative summary of the evaluation, focusing on the following aspects:

Abbreviation	Benchmark Description
MMLU	Massive Multitask Language Understanding: Evaluates the model's knowledge and skills across a wide variety of tasks.
IFEval	Evaluation of the model's ability to follow instructions effectively.
HumanEval	Evaluation of the model's ability to generate correct code from problem descriptions.
MBPP EvalPlus	Benchmark to evaluate the model's capability in generating code, especially in more advanced programming problems.
GSM8K	Benchmark that evaluates the model's mathematical reasoning.
MATH	Benchmark focused on assessing the model's ability to solve advanced-level mathematical problems.
ARC Challenge	Benchmark measuring the model's scientific reasoning and world knowledge capabilities.
GPQA	General Purpose Question Answering: Evaluation of the model's ability to answer general-purpose questions.
BFCL	Evaluation of the model in the use of advanced tools and functionalities.
Nexus	Benchmark measuring the model's ability to use tools and technologies in complex scenarios.
ZeroSCROLLS/QuALITY	Evaluation of the model's ability to understand and process long contexts.
InfiniteBench/En.MC	Benchmark measuring the model's ability to handle extremely long context sequences.
NIH/Multi-needle	Evaluation of the use of tools and capabilities in medical and scientific environments.
MGSM	Multilingual benchmark evaluating the model's ability to reason and solve problems in multiple languages.

Modelo/Benchmark	ARC Challenge (0-shot)	GPQA (0-shot, CoT)	BFCL	Nexus	ZeroSCROLLS/QuALITY	InfiniteBench/En.MC	NIH/Multi-needle	MGSM (0-shot, CoT)
Llama 3 (8B)	83,40%	32,80%	76,10%	38,50%	81,00%	65,10%	98,80%	68,90%
Llama 3 (70B)	94,80%	46,70%	84,80%	56,70%	90,50%	78,20%	97,50%	86,90%
Llama 3 (405B)	96,90%	51,10%	88,50%	58,70%	95,20%	83,40%	98,10%	91,60%
GPT-3.5 Turbo	83,70%	30,80%	85,90%	37,20%	-	-	-	51,40%
GPT-4	96,70%	53,60%	90,20%	56,10%	90,50%	82,50%	100,00%	90,50%
Mistral 7B	74,20%	28,80%	60,40%	24,70%	-	-	-	29,90%
Mixtral 8x22B	88,70%	33,30%	-	48,50%	-	-	-	71,10%
Claude 3.5 Sonnet	96,70%	59,40%	93,10%	45,70%	90,50%	-	100,00%	91,60%
Gemma 2 (9B)	87,60%	-	-	30,00%	-	-	-	53,20%
Gemini 1.5 Pro	92,30%	46,20%	88,50%	58,70%	-	-	99,70%	75,20%
Gemini 1.5 Flash	-	-	-	-	-	-	99,10%	71,70%
Falcon2-11B	59,73%	-	-	-	-	-	-	-
Nemotron 4 (340B)	94,60%	-	86,50%	-	-	-	-	-
Grok-1.5 *		35,90%						
Grok-2 mini‡ *		51,00%						
Grok-2‡ *		56,00%						

Modelo/Benchmark	MMLU (5-shot)	MLU (0-shot, Co	MMLU-Pro (5-shot, CoT)	IFEval	HumanEval (0-shot)	MBPP EvalPlus (0-shot)	GSM8K (8-shot, CoT)	MATH (0-shot, CoT)
Llama 3 (8B)	66,60%	73,00%	48,30%	80,40%	72,60%	72,80%	84,50%	51,90%
Llama 3 (70B)	79,20%	86,00%	66,40%	87,50%	80,50%	86,00%	95,10%	68,00%
Llama 3 (405B)	87,30%	88,60%	73,30%	88,60%	89,00%	88,60%	96,80%	73,80%
GPT-3.5 Turbo	70,70%	69,80%	49,20%	69,90%	68,00%	82,00%	81,60%	43,10%
GPT-4	89,10%	88,70%	74,00%	85,60%	90,20%	87,80%	96,10%	76,60%
Mistral 7B	62,50%	60,50%	36,90%	57,60%	40,20%	49,50%	53,20%	13,00%
Mixtral 8x22B	76,90%	79,90%	56,30%	72,70%	75,60%	78,60%	88,20%	54,10%
Claude 3.5 Sonnet	90,40%	88,30%	77,00%	88,00%	92,00%	90,50%	96,40%	71,10%
Gemma 2 (9B)	72,30%	72,34%	-	73,60%	54,30%	71,70%	76,70%	44,30%
Gemini 1.5 Pro	85,90%	84,10%	75,30%	82,60%	84,10%	82,60%	94,10%	67,70%
Gemini 1.5 Flash	82,00%	-	-	-	-	-	-	63,90%
Falcon2-11B	58,37%	-	82,07%	-	29,60%	-	53,83%	-
Nemotron 4 (340B)	82,60%	78,70%	62,70%	85,10%	73,20%	72,80%	92,30%	41,10%
Grok-1.5 *	81,30%		51,00%		74,10%			50,60%
Grok-2 mini‡ *	86,20%		72,00%		85,70%			73,00%
Grok-2‡ *	87,50%		75,50%		88,40%			76,10%

For each benchmark, there is a range from red to green (with green being the best score and red the worst).

7. Required Infrastructure

7.1.1. LLM Deployment

For testing the LLM systems, they have been deployed either locally or remotely.

The annex "20240925_COGNITREK_Analysis_ANNEX-LLMs-ES.pdf" includes the various deployments carried out.

7.2. XD1R Module

User management via JWT with a Permission System Hashed through Session Cookies by the Backend. Due to the need to separate sensitive information, two distinct databases will be used.

Minimum server requirements:

- RAM: 8gb
- CPU: 4 kernels, 3.0 ghz
- CPU: (basic)
- HDD: 250-500GB
- SO: Linux / ubuntu 20-22
- PHP 8.2
- BACKEND: Laravel 9.
- DDBB: with 2 separate drivers for data:
  - User Model → Sensitive Information
  - Other Model → Non senstive information.
- Node JS 20
- Tests: PHP Unit test
- API: Swagger
- RestFull API
- Caching: Redis

FRONTEND: ReactJS (Vite, NextJS) con gestor de estado global (Redux)

7.3. XDAM Module

Minimum server requirements:

- RAM: 8gb
- CPU: 4 kernels, 3.0 ghz
- CPU: (basic)
- HDD: 1 Tbyte.
- SO: Linux / ubuntu 20-22
- PHP 8.2
- BACKEND: Laravel 9.
- DDBB: Mysql or MariaDB
- Apache SOLR 9.
- Node JS 20
- Tests: PHP Unit test
- API: Swagger
- RestFull API
- Caching: Redis

FRONTEND: ReactJS (with Redux)

8. References

- FALCON2-11B TECHNICAL REPORT, Technology Innovation Institute, Abu Dhabi, United Arab Emirates, 2024.
- Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context, Google 2024.
- Gemma 2: Improving Open Language Models at a Practical Size, Google 2024.
- GPT-4o: Visual perception performance of multimodal large language models in piglet activity understanding, 2024.
- Mixtral of Experts, 2024.
- The Llama 3 Herd of Models, META 2024.
- Claude 3.5 Sonnet Model Card Addendum, Anthropic 2024.